Evaluating Vision-Based Human Motion Prediction for Safe Robot Navigation

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***Abstract*—This paper evaluates monocular RGB-based human motion estimation and short-term trajectory prediction for safe robot navigation in human-populated environments. A single participant performed natural locomotion in a controlled indoor lab while simultaneously captured by a monocular RGB camera and an OptiTrack motion capture system at 120 Hz and 125 Hz respectively. YOLOv8-Pose extracted 2D keypoints geometrically lifted to 3D using a Kalman Filter and compared against synchronized OptiTrack ground truth. Position estimation error was quantified using RMSE, MAE, and Pearson correlation across horizontal, vertical, and depth axes. Trajectory prediction was evaluated at 0.5s, 1.0s, and 2.0s horizons. Results show strong horizontal tracking (RMSE = 118.4 mm, r = 0.962) but significant depth estimation failure (RMSE = 711.5 mm), propagating into a 65% false positive collision rate — rendering the system impractical despite zero missed dangers. These findings provide a quantitative basis for assessing low-cost monocular vision as an alternative to depth sensors for human-aware robot navigation.**

***Index Terms*—Human motion estimation, monocular RGB camera, robot navigation, motion capture, OptiTrack, pose estimation, trajectory prediction, Kalman filter, human–robot interaction**

I. Introduction

Autonomous robot navigation in environments shared with humans such as hospitals, warehouses, and indoor service settings requires accurate and reliable perception of human motion. Robots operating in these spaces must continuously estimate and anticipate human movement to avoid collisions, maintain safe distances, and comply with social norms [1], [2]. The ability to perceive and predict short-term human trajectories is therefore a critical component of any human-aware navigation system, and errors in this perception pipeline can directly compromise safety [5].

Vision-based approaches have become increasingly popular for human motion estimation due to their flexibility and relatively low cost compared to specialized sensing hardware. RGB cameras in particular offer a practical sensing modality, capable of capturing rich spatial and temporal information about human pose and movement without requiring physical markers or structured environments [4], [7]. Recent work has demonstrated that pose estimation pipelines applied to RGB video can extract meaningful motion trajectories in real time, enabling downstream applications such as trajectory prediction and collision risk assessment [5], [7].

However, monocular RGB perception introduces inherent limitations that are not present in depth-enabled systems. Unlike RGB-D cameras, monocular RGB sensors cannot directly measure scene depth, leading to ambiguity in three-dimensional position estimation [6]. Additionally, factors such as occlusion, lighting variation, and motion blur can degrade tracking quality and introduce noise into estimated trajectories. These inaccuracies can propagate into navigation safety decisions such as minimum human–robot distance estimation and collision risk prediction potentially compromising safety in dynamic environments [5], [7].

A common approach in the literature is to use RGB-D cameras, such as the Intel RealSense or Stereo ZED 2, which provide both color and depth information and therefore reduce the ambiguity associated with monocular systems [8]. However, RGB-D sensors are more expensive, power intensive, and sensitive to environmental conditions such as strong lighting or reflective surfaces, which limits their practicality in some deployment scenarios. Monocular RGB cameras, by contrast, are low-cost, lightweight, and widely available, making them attractive for deployment on mobile platforms where size and budget are constrained [6].

Despite growing interest in vision-based human motion estimation for robotics, limited work has systematically evaluated the accuracy of monocular RGB-based approaches against high-precision ground truth under synchronized conditions. Many prior studies either rely on depth-enabled sensors or perform only qualitative comparisons due to challenges in data synchronization [9], [10]. This gap motivates the present work.

This paper presents a study evaluating the accuracy of monocular RGB-based human motion estimation and short-term trajectory prediction for safe robot navigation. Human motion is captured simultaneously using an RGB camera and an OptiTrack optical motion capture system, providing temporally synchronized vision-based estimates and high-precision ground-truth trajectories. The contributions of this work are as follows:

- A synchronized RGB and motion capture dataset collected under controlled indoor conditions simulating a human–robot interaction scenario.
- A quantitative analysis of monocular RGB-based pose estimation accuracy against OptiTrack ground-truth trajectories across three spatial axes.

- A Kalman Filter-based trajectory prediction evaluation at prediction horizons of 0.5 s, 1.0 s, and 2.0 s, compared against OptiTrack ground truth.
- An assessment of how RGB perception limitations affect safety-critical navigation metrics including minimum human–robot distance and collision risk.

II. Related Work

A. Human-Aware Robot Navigation

Human-aware robot navigation has received significant attention in recent years, particularly in environments with high human activity such as hospitals and service settings. These approaches aim to improve both safety and social compliance by incorporating models of human motion into the robot's planning process. Prior work has explored predicting human trajectories to enable robots to anticipate future movements and adjust their paths accordingly. Methods based on long-term motion prediction and behavior modeling have been proposed to enhance navigation efficiency and reduce collision risk [5], [7]. Other studies have focused on patient–robot co-navigation in crowded environments, emphasizing the importance of predicting human intent and maintaining safe distances [3]. Eirale et al. provide a comprehensive review of human following and guidance strategies for autonomous mobile robots, highlighting the increasing role of perception accuracy in safe navigation [4]. These works collectively demonstrate that accurate human motion estimation is a fundamental prerequisite for socially aware and safe robot navigation.

B. Vision-Based Human Motion Estimation

Vision-based approaches, particularly those using RGB and RGB-D cameras, have become widely adopted for human motion estimation due to their ability to provide rich spatial and temporal information. Recent surveys have reviewed techniques for extracting human pose, estimating trajectories, and predicting motion from visual data [8]. Rodriguez-Martinez et al. highlight the growing role of vision-based systems in autonomous robot navigation and perception, noting both the promise and the practical challenges of deploying such systems in unstructured environments [6]. However, these methods are often affected by challenges such as sensor noise, occlusions, and in the case of monocular RGB systems depth ambiguity, which can degrade tracking performance and introduce uncertainty in motion estimation [6], [8]. Understanding these limitations is essential for evaluating the reliability of vision-based perception systems in safety-critical applications. Pose estimation frameworks such as YOLOv8-Pose have enabled real-time 2D keypoint detection from RGB video, forming the basis of many human motion estimation pipelines [4]. While these tools are efficient and accessible, translating 2D keypoints into accurate 3D motion trajectories remains a challenge for monocular systems, often requiring additional assumptions or learned depth priors [6], [8].

C. Motion Capture as Ground Truth Reference

Motion capture systems, such as OptiTrack, are widely used as high-precision ground-truth references for evaluating vision-based human motion estimation. These systems rely on reflective markers and multiple synchronized cameras to capture highly accurate spatial trajectories at high sampling rates. Prior research has utilized motion capture data to validate markerless tracking systems and assess the accuracy of vision-based approaches [9], [10]. By providing reliable reference trajectories, motion capture systems enable detailed analysis of positional and temporal errors in estimated motion. Bajrami et al. conducted a comparative evaluation of multi-view markerless upper-limb motion tracking approaches, demonstrating the value of motion capture as a benchmark for assessing RGB-based estimation accuracy [12]. In this work, OptiTrack data is used as a synchronized ground-truth reference for direct comparison with RGB-based motion estimates.

III. Methodology

A. Experimental Setup

The data collection was conducted in a dedicated motion capture laboratory equipped with a ceiling-mounted truss system supporting multiple sensors. The experimental space simulates an indoor human-populated environment representative of healthcare and service robotics deployment scenarios, where safe human–robot interaction is a critical requirement.

The laboratory is instrumented with an OptiTrack motion capture system consisting of multiple Prime Color infrared cameras mounted on the ceiling truss at various positions and angles surrounding the capture volume. The cameras are positioned to provide overlapping coverage of the full capture area, ensuring continuous tracking of reflective markers throughout all participant movements.

In addition to the OptiTrack system, a monocular RGB camera is mounted on the ceiling structure and angled downward to capture an overhead view of the participant within the scene. This camera serves as the primary vision-based sensing modality in this study, simulating the perspective of a robot-mounted or environment-mounted RGB camera observing human motion from an elevated fixed position. Both systems were operated simultaneously during all data collection sessions at 120 Hz and 125 Hz respectively, ensuring temporal synchronization between the RGB video stream and the OptiTrack motion capture data.

B. Participant and Marker Configuration

A single participant was recruited for data collection and equipped with individual reflective markers placed at key anatomical locations across the body. Specifically, markers were positioned on the head (3 markers), each hand (3 markers per hand), and each knee (2 markers per knee), for a total of 13 markers. This configuration enables the OptiTrack system to capture accurate three-dimensional trajectories for the head and limb segments most relevant to human–robot interaction analysis.

C. Data Collection

Data collection consisted of a series of recorded trials in which the participant performed natural human motion within the capture volume. The participant was instructed to perform representative activities including walking across the scene, turning, pausing, and performing natural upper-body movements such as arm gestures and reaching motions.

The OptiTrack system recorded 3D marker trajectories at approximately 125 Hz throughout each trial, providing high-frequency, high-precision spatial data. The RGB camera recorded video concurrently at approximately 120 Hz. The total recorded dataset spans 80 seconds of continuous motion, covering a range of motion speeds and directions.

D. Vision-Based Motion Estimation Pipeline

Human motion is estimated from the RGB video using a YOLOv8-Pose estimation pipeline applied frame-by-frame. For each frame, 2D keypoints corresponding to major body landmarks including the nose, shoulders, hips, and knees are detected and tracked over time. A center-crop preprocessing step is applied to the RGB frames prior to inference to isolate the participant from other individuals in the scene.

The nose keypoint is used as the primary head position proxy, as it most directly corresponds to the OptiTrack head marker position. The body center trajectory is additionally computed as the mean pixel coordinate of the left and right shoulder and hip keypoints:

$$x_{\text{body}} = \tfrac{1}{4}(x_{ls} + x_{rs} + x_{lh} + x_{rh}) \tag{1}$$

where x_{ls}, x_{rs}, x_{lh}, and x_{rh} denote the pixel x-coordinates of the left shoulder, right shoulder, left hip, and right hip, respectively.

Since the monocular RGB camera does not provide direct depth information, 3D position is estimated geometrically using the vertical span between the nose and knee keypoints in the image plane. The raw depth estimate is computed as:

$$Z_{\text{rgb}} = \frac{f_x \cdot H_{\text{real}}}{h_{\text{pixel}}} \tag{2}$$

where f_x is the approximate focal length derived from the camera resolution and assumed field of view, $H_{\text{real}} = 900\,\text{mm}$ is the assumed real-world nose-to-knee height, and h_{pixel} is the pixel height of the person computed as the vertical distance between the nose and knee midpoint keypoints. A metric scale factor is subsequently derived from the OptiTrack ground-truth depth values using the median ratio between the two signals.

Temporal alignment between the RGB-based estimates and OptiTrack ground-truth trajectories is achieved using synchronized timestamps. For each RGB frame, the closest OptiTrack frame within a 50 ms tolerance window is matched using a nearest-neighbor timestamp join.

E. Ground-Truth Motion Capture

The OptiTrack system serves as the ground-truth reference. The three head markers are averaged per-frame to produce a single clean 3D head trajectory:

$$\mathbf{p}_{\text{head}} = \frac{1}{3}\sum_{k=1}^{3} \mathbf{m}_k \tag{3}$$

where $\mathbf{m}_k = (X_k, Y_k, Z_k)$ denotes the 3D position of the k-th head marker. The X component represents lateral position, the Y component represents height, and the Z component represents depth.

F. Kalman Filter-Based Trajectory Prediction

A one-dimensional Kalman Filter is applied independently to each spatial axis of the RGB-estimated position to smooth sensor noise and enable short-term trajectory prediction. The state vector is defined as $\mathbf{x} = [p, v]^\top$, representing position and velocity. The state transition and measurement models are:

$$\mathbf{F} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} 1 & 0 \end{bmatrix} \tag{4}$$

The KF is applied exclusively to the RGB-estimated positions. The OptiTrack system is used only as ground truth and is not processed by the KF. Predictions are made at horizons of $h \in \{0.5, 1.0, 2.0\}$ s and evaluated against the OptiTrack ground-truth position at the corresponding future timestep.

IV. Evaluation Metrics and Error Analysis

A. Positional Estimation Error

Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) are computed between the RGB-estimated position and the OptiTrack ground-truth position for each spatial axis:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{t=1}^{N}\left(\hat{p}_t - p_t^{GT}\right)^2} \tag{5}$$

$$\text{MAE} = \frac{1}{N}\sum_{t=1}^{N}\left|\hat{p}_t - p_t^{GT}\right| \tag{6}$$

The Pearson correlation coefficient r is additionally computed to quantify the degree of trajectory shape agreement between the RGB estimate and OptiTrack ground truth.

B. Trajectory Prediction Error

Prediction RMSE and MAE are computed between the KF-predicted future position (from RGB input) and the OptiTrack ground-truth position at the same future timestep, for each prediction horizon h.

C. Depth Ambiguity Quantification

The depth dimension of the OptiTrack trajectory (Z coordinate) is analyzed to quantify the magnitude of depth movement unobservable to the monocular RGB camera:

$$\Delta Z = Z_{\max} - Z_{\min} \tag{7}$$

D. Human–Robot Distance Error

A robot is assumed stationary at $(X_r, Z_r) = (800, -2000)$ mm. The true human–robot distance is:

$$d_{\text{real}}(t) = \sqrt{(X(t) - X_r)^2 + (Z(t) - Z_r)^2} \quad (8)$$

The RGB-estimated distance uses only lateral information:

$$d_{\text{RGB}}(t) = |x_{\text{body,norm}}(t) \cdot (X_{\max} - X_{\min}) + X_{\min} - X_r| \quad (9)$$

The distance estimation error is $\epsilon_d(t) = |d_{\text{real}}(t) - d_{\text{RGB}}(t)|$.

E. Collision Risk Classification

A safety threshold of $d_{\text{safe}} = 500$ mm is defined. False negatives occur when $d_{\text{real}} < d_{\text{safe}}$ but $d_{\text{RGB}} \geq d_{\text{safe}}$ (missed danger). False positives occur when $d_{\text{RGB}} < d_{\text{safe}}$ but $d_{\text{real}} \geq d_{\text{safe}}$ (unnecessary stop).

V. Results

A. Dataset Summary

The paired dataset comprises 8,242 synchronized frames spanning 80 seconds of continuous motion (5–85 s). After filtering low-confidence detections (confidence < 0.1), OptiTrack occlusion frames (head height $< 1{,}650$ mm), and RGB keypoint spikes (frame-to-frame jump > 80 px), 5,097 valid frames were retained (61.8% of the total dataset).

B. Position Estimation Error

Fig. 1 shows the estimated trajectories from both systems across all three spatial axes. Table I summarizes the estimation error metrics.

TABLE I
Position Estimation Error: RGB Camera vs OptiTrack Ground Truth

Axis	RMSE (mm)	MAE (mm)	Correlation
Horizontal Position	118.4	84.7	0.962
Vertical Position	19.1	14.4	0.323
Distance from Camera	711.5	473.4	0.767

The RGB camera achieves strong horizontal position tracking, with an RMSE of 118.4 mm and a Pearson correlation of $r = 0.962$. Vertical position estimation is the most accurate axis (RMSE = 19.1 mm), reflecting the stability of the head height during walking. Distance from camera (depth) shows the largest estimation error (RMSE = 711.5 mm, $r = 0.767$), consistent with the known limitations of monocular depth estimation.

C. Estimation Error Over Time

Fig. 2 shows the per-frame estimation error for each spatial axis over the recording duration. Horizontal error is bounded within approximately ± 600 mm, with larger errors occurring at trajectory reversal points where the RGB estimate lags the ground truth. Vertical error is consistently small (< 70 mm). Depth error shows pronounced spikes at $t \approx 20$ s and $t \approx 33$ s,

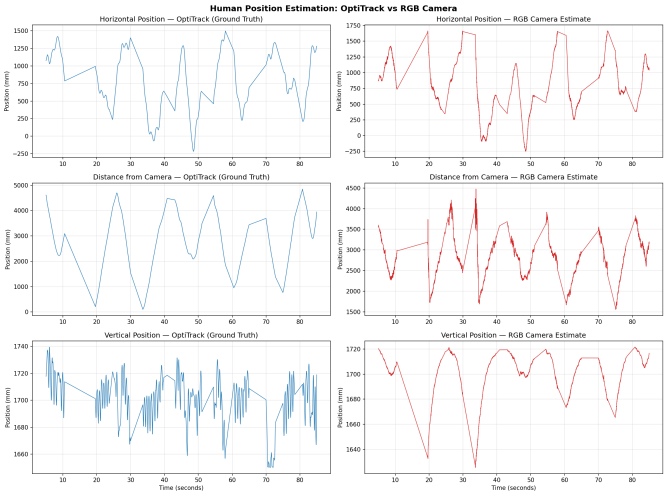


Fig. 1. Human position estimation comparison: OptiTrack ground truth (blue) vs RGB camera estimate (red) for horizontal position, distance from camera, and vertical position.

corresponding to close-proximity events where the participant approached the camera. These spikes indicate that geometric depth estimation degrades significantly at close range – a critical finding for robot navigation safety.

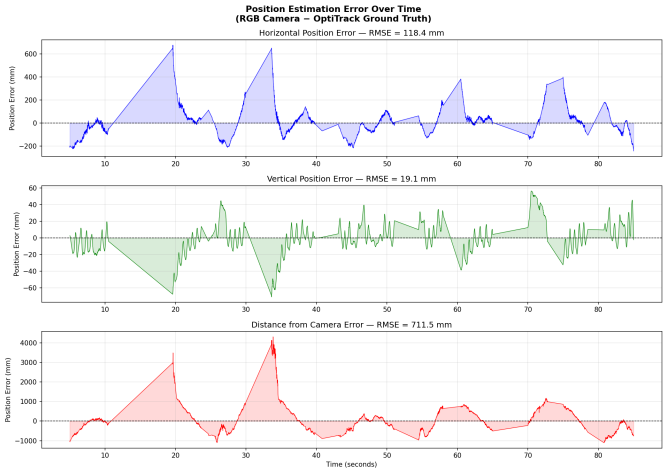


Fig. 2. Position estimation error over time (RGB − OptiTrack ground truth) for all three spatial axes. Shaded regions indicate signed error magnitude.

D. Depth Ambiguity

Fig. 3 compares the depth component of the OptiTrack ground-truth trajectory against the RGB camera depth estimate. The participant traversed a total depth range of $\Delta Z = 5{,}256$ mm (approximately 5.3 m) across the 80-second recording, with a standard deviation of 1,380 mm. While the RGB depth estimate correctly identifies the general pattern of depth variation ($r = 0.767$), it consistently underestimates extreme close distances. The RGB estimate remained bounded between 1,559 mm and 4,475 mm, failing to capture the full dynamic range of the OptiTrack trajectory (100 mm to 5,000 mm).

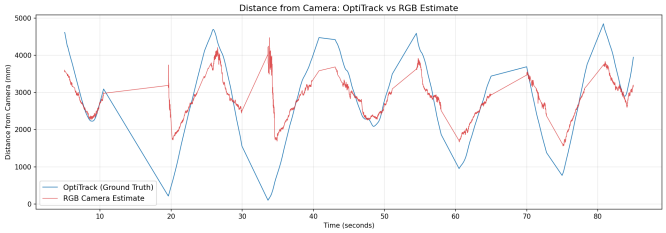


Fig. 3. Distance from camera: OptiTrack ground truth (blue) versus RGB geometric depth estimate (red). RGB consistently underestimates close-range depth.

E. Kalman Filter Trajectory Prediction

Table II summarizes the KF-RGB prediction error evaluated against OptiTrack ground truth at three prediction horizons.

TABLE II
KF-RGB TRAJECTORY PREDICTION ERROR VS OPTITRACK GROUND TRUTH

Axis	0.5 s		1.0 s		2.0 s	
	RMSE (mm)	MAE (mm)	RMSE (mm)	MAE (mm)	RMSE (mm)	MAE (mm)
Horizontal	246	176	389	303	719	586
Vertical	23	16	27	18	36	25
Depth	810	472	1005	693	1529	1308

Fig. 4 illustrates how prediction error grows with the prediction horizon for all three axes. Horizontal prediction error grows from 246 mm at 0.5 s to 719 mm at 2.0 s, while vertical prediction error remains small and stable (23–36 mm across all horizons). Depth prediction error is the most critical, exceeding 1,000 mm at a 1.0 s horizon – representing the distance within which robot collision avoidance decisions must be made.

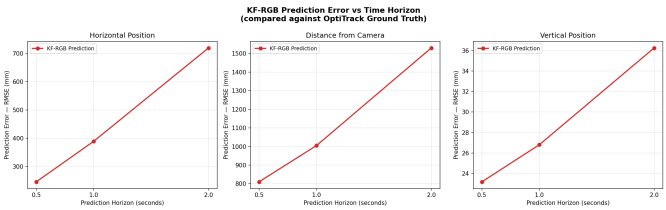


Fig. 4. KF-RGB prediction error (RMSE) vs prediction horizon (0.5 s, 1.0 s, 2.0 s) for horizontal position, distance from camera, and vertical position. All values evaluated against OptiTrack ground truth.

Fig. 5 shows the KF-predicted trajectories overlaid against the OptiTrack ground truth at a 1.0 s prediction horizon. The horizontal prediction closely follows the ground-truth motion pattern, while depth prediction shows persistent offset errors, particularly during close-proximity events.

F. Human–Robot Distance Error

The true human–robot distance ranged from 113 mm to 3,542 mm with a mean of 1,586 mm. The RGB-estimated distance ranged from near zero to approximately 1,100 mm with a mean of 397 mm, indicating a systematic underestimation of the true distance. The distance estimation error had a MAE of 1,198 mm and an RMSE of 1,502 mm – both exceeding the 500 mm safety threshold.

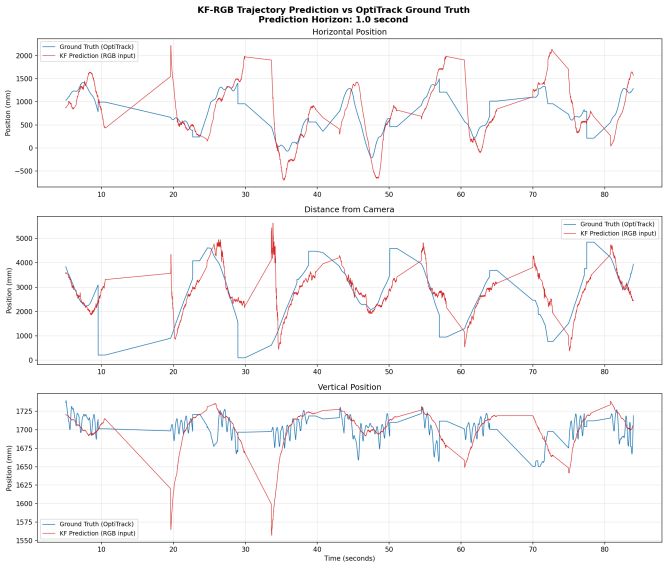


Fig. 5. KF-RGB trajectory prediction (red) vs OptiTrack ground truth (blue) at a 1.0 s prediction horizon for all three spatial axes.

G. Collision Risk Analysis

Table III summarizes the collision risk classification results.

TABLE III
COLLISION RISK CLASSIFICATION RESULTS

Metric	Value	Implication
True unsafe (GT)	5.1%	Real danger frames
RGB predicted unsafe	70.1%	RGB safety estimate
False negatives	0.0%	Missed danger events
False positives	65.0%	Unnecessary stops

The RGB system predicted an unsafe condition in 70.1% of all frames – a 13.7× overestimate relative to the 5.1% ground-truth unsafe rate. While the false-negative rate was 0.0% (no missed danger events), the false-positive rate of 65.0% means the robot would halt unnecessarily in nearly two-thirds of all frames, rendering the system practically unusable.

H. Summary of Results

Table IV consolidates all quantitative results.

VI. DISCUSSION

A. Horizontal Tracking Reliability

The strong horizontal position correlation ($r = 0.962$, RMSE $= 118.4$ mm) confirms that monocular RGB perception reliably captures lateral human motion from an overhead camera perspective. This is consistent with prior work showing that 2D keypoint-based lateral tracking performs well when the dominant motion is within the camera's image plane [4], [6].

TABLE IV
SUMMARY OF ALL EVALUATION RESULTS

Metric	Value	Unit
Horizontal estimation RMSE	118.4	mm
Vertical estimation RMSE	19.1	mm
Depth estimation RMSE	711.5	mm
Horizontal correlation (r)	0.962	—
Depth correlation (r)	0.767	—
Depth range unobserved	5,256	mm
KF prediction RMSE (horiz, 1.0 s)	389	mm
KF prediction RMSE (depth, 1.0 s)	1,005	mm
KF prediction RMSE (vert, 1.0 s)	27	mm
Distance MAE	1,198	mm
Distance RMSE	1,502	mm
False negatives	0.0	%
False positives	65.0	%

The Kalman Filter further demonstrates that this lateral signal can support short-term prediction with acceptable accuracy at a 1.0 s horizon (RMSE $= 389$ mm), suggesting that horizontal trajectory prediction from monocular RGB is viable for robot navigation planning in lateral avoidance scenarios.

B. Depth Ambiguity as the Critical Limitation

The most significant finding of this study is the magnitude of depth estimation error and its propagation into navigation safety metrics. The participant traversed over 5 m in depth during the recording, yet the RGB geometric depth estimate remained bounded between 1,559 mm and 4,475 mm – failing to capture extreme close-range or far-range positions. At a 1.0 s prediction horizon, the KF-RGB depth prediction error reaches 1,005 mm, exceeding the standard 500 mm personal space boundary used in human–robot interaction research [5]. This confirms that depth ambiguity is not merely a theoretical concern but a practical safety risk in real navigation scenarios. Consistent with Machkour et al. [13], monocular depth estimation degrades significantly at close range – confirming this is a fundamental limitation of monocular sensing regardless of method.

The depth estimation error also shows strong distance-dependency: error spikes at $t \approx 20$ s and $t \approx 33$ s correspond to close-proximity events where the geometric depth model breaks down. This nonlinear error behavior is particularly dangerous for robot navigation, as close-proximity errors are precisely when accurate depth estimation is most critical.

C. Kalman Filter Effectiveness

The Kalman Filter proved effective at stabilizing noisy RGB position estimates for trajectory prediction. By maintaining an internal state estimate of position and velocity, the KF produces smooth predictions that resist frame-to-frame noise in keypoint detections. This is particularly important for depth prediction, where raw RGB depth estimates are highly noisy due to the approximate nature of geometric depth estimation from pixel heights. The KF reduces depth prediction instability and produces more interpretable trajectory forecasts, making it a practical choice for real-time robot navigation systems with noisy monocular perception inputs.

D. Collision Risk Implications

The collision risk analysis reveals a fundamental asymmetry in the error behavior of monocular RGB for safety applications. The 0.0% false-negative rate – meaning the RGB system never missed a genuine unsafe proximity event – appears beneficial but is actually a consequence of systematic distance underestimation. Because the RGB estimate collapses 3D distance to a 1D lateral offset, it consistently underestimates the true human–robot distance, which causes it to trigger safety responses earlier than necessary. The resulting 65.0% false-positive rate means the robot would halt in approximately two-thirds of all operational frames, making the system impractical despite its apparent safety guarantees. This finding highlights that safety metrics for monocular RGB systems should not be evaluated solely on false-negative rates but must account for operational efficiency and false-positive burden.

E. Implications for Robot Navigation System Design

The results collectively suggest a layered sensor strategy for safe robot navigation. Monocular RGB is sufficient for lateral trajectory estimation and prediction, and could serve as a primary modality for horizontal obstacle avoidance. However, depth sensing – whether through RGB-D cameras, LiDAR, stereo vision, or learned monocular depth estimation – is essential for reliable distance-based safety metrics and collision prediction. A practical architecture would combine monocular RGB for fast, low-cost lateral tracking with a supplementary depth source for distance estimation and close-range safety monitoring.

VII. LIMITATIONS

Several limitations should be acknowledged when interpreting the findings of this study.

Single participant. The dataset was collected from a single participant performing scripted activities in a controlled laboratory. The results may not generalize to diverse body types, motion styles, or multiple simultaneous agents. Future work should replicate the evaluation with multiple participants across varied motion patterns.

Approximate camera intrinsics. Camera intrinsic parameters (focal length, principal point) were approximated from image resolution and an assumed field of view of 65°, as the exact calibration of the ceiling-mounted RGB camera was unavailable. This approximation introduces systematic error into the geometric depth estimation pipeline and affects the absolute accuracy of the depth RMSE metric. Proper checkerboard calibration would improve the reliability of depth estimates.

Geometric depth estimation. The depth estimation method relies on a fixed anthropometric prior ($H_{\text{real}} = 900$ mm nose-to-knee) and geometric projection. This model assumes a static, upright participant and does not account for body tilt, crouching, or occluded keypoints. More accurate depth could be obtained using learned monocular depth estimation models such as Depth Anything V2 [8], which would be a natural extension of this work.

Coordinate system alignment. The transformation between the OptiTrack world coordinate frame and the RGB camera coordinate frame was achieved through statistical normalization rather than a formal extrinsic calibration. A proper rigid-body transformation derived from calibration targets would improve alignment accuracy and enable more direct metric comparison.

Single static robot position. The human–robot distance analysis assumes a single, stationary robot at a fixed position in the capture volume. Real robot navigation involves dynamic robot trajectories, and the distance error characteristics may differ under moving robot conditions.

Controlled environment. The experiment was conducted in a controlled laboratory under stable lighting conditions. Performance may degrade in real-world deployment environments with variable lighting, background clutter, or multiple people in the scene.

Prediction model scope. Only a Kalman Filter-based prediction model was evaluated. More expressive models such as Long Short-Term Memory (LSTM) networks or Transformer-based trajectory predictors [7] may achieve lower prediction error, particularly at longer horizons, and represent a natural direction for future investigation.

VIII. Conclusion

This paper presented a quantitative evaluation of monocular RGB-based human motion estimation and short-term trajectory prediction for safe robot navigation, using a synchronized OptiTrack motion capture system as high-precision ground truth. The study examined position estimation error across three spatial axes, Kalman Filter-based prediction error at multiple time horizons, human–robot distance estimation error, and collision risk classification across 80 seconds of synchronized motion data.

The results demonstrate that monocular RGB reliably captures horizontal human motion ($r = 0.962$, RMSE $= 118.4$ mm) and that Kalman Filter prediction achieves acceptable horizontal trajectory prediction (RMSE $= 389$ mm at 1.0 s). However, depth estimation introduces substantial error (RMSE $= 711.5$ mm), which propagates into prediction error exceeding 1,000 mm at a 1.0 s horizon and human–robot distance errors of 1,198 mm MAE – more than twice the 500 mm safety threshold. The consequence is a 65.0% false-positive rate for collision detection, rendering the system impractical for real deployment despite its zero false-negative rate.

These findings confirm that monocular RGB sensing alone is insufficient for reliable safety-critical robot navigation in dynamic indoor environments. Practical deployments should supplement monocular RGB with depth sensing or sensor fusion approaches. Future work should investigate learned monocular depth estimation to reduce depth errors, extend the evaluation to multiple participants and dynamic robot trajectories, and explore more expressive prediction models for longer-horizon trajectory forecasting.

References

[1] K. Karwowski et al., "A survey on socially aware robot navigation: Taxonomy and future challenges," *International Journal of Robotics Research*, 2024.

[2] E. Stefanini et al., "Efficient context-aware model predictive control for human-aware navigation," *IEEE Robotics and Automation Letters*, vol. 9, no. 11, 2024.

[3] K. Kodur and M. Kyrarini, "Patient–robot co-navigation of crowded hospital environments," *Applied Sciences*, vol. 13, no. 7, 2023.

[4] A. Eirale, M. Martini, and M. Chiaberge, "Human following and guidance by autonomous mobile robots: A comprehensive review," *IEEE Access*, 2025.

[5] L. Bruckschen, K. Bungert, N. Dengler, and M. Bennewitz, "Human-aware robot navigation by long-term movement prediction," in *Proc. IEEE/RSJ IROS*, 2020, pp. 11032–11037.

[6] E. A. Rodriguez-Martinez et al., "Vision-based navigation and perception for autonomous robots," *Eng*, vol. 6, no. 7, 2025.

[7] Z. Zhang et al., "Future-oriented navigation: Dynamic obstacle avoidance with one-shot energy-based multimodal motion prediction," *IEEE Robotics and Automation Letters*, 2025.

[8] H. Liu and H. Wang, "A review on vision-based motion estimation," *arXiv:2407.14478*, 2024.

[9] B. Castillo et al., "Assessing spatiotemporal behavior of human gait," *Experimental Techniques*, vol. 49, no. 1, pp. 3–13, 2025.

[10] M. Hatamzadeh, "Motion analysis by video for gait evaluation with innovative technology," Ph.D. dissertation, Universite Cote d'Azur, 2024.

[11] C. Y. Chang et al., "Towards pervasive physical rehabilitation using Microsoft Kinect," in *Proc. PervasiveHealth*, 2012, pp. 159–162.

[12] A. Bajrami et al., "Markerless upper limb motion tracking: A comparative evaluation of multi-view approaches," in *Proc. IEEE MetroXRAINE*, 2025, pp. 818–823.

[13] Z. Machkour, D. Ortiz-Arroyo, and P. Durdevic, "Monocular based navigation system for autonomous ground robots using multiple deep learning models," *International Journal of Computational Intelligence Systems*, vol. 16, no. 79, 2023.